

External costs of high-voltage overhead transmission lines: Evidence from a causal approach in rural France

Abstract

The electricity transmission network needs to be developed for the integration of renewable sources. In North America and Europe, to achieve renewable electricity goal, massive investments in electricity transport infrastructure are planned to connect solar and wind generation sites to the grid network. However, local oppositions to new installations of high voltage overhead transmission lines (HVOTLs) are causing delays and overrun costs. Visual nuisance, is an important argument for local stakeholders' opposition to a new project. Estimations of the local economic impact of HVOTLs help in identifying the external costs of the electricity supply industry to improve both efficiency in electricity pricing and fairness in defining host community compensation provision. Prior research has provided evidence that existing HVOTLs have negative but not substantial effects to their immediate neighbourhoods. However, these studies have applied correlational approach. This paper revisits the external effects of HVOTLs exploring the case of a 225 kV line connecting two medium cities (Saint-Privat-d'Allier to Saint-Etienne) in France. We take advantage of the public disclosure in 2009 of undergrounding 8 km out of a total 87 km lines as a result of a network upgrade project, to develop a difference-in-difference specification of a hedonic regression model, by comparing house sale prices at different distances from existing pylons, before and after the announcement. The announcement of undergrounding materializes by a rise of about 40% in the price of treated houses (located within 200 m of old pylons). The implied environmental costs of proximity to high voltage pylons are substantial and their reduction is internalized into house prices as soon as the decision to underground is announced. These results are robust to alternative specifications.

Key words:

Transmission Line; Expectations; Hedonic valuation; Difference-in-Differences Estimator; Policy Analysis.

External costs of high-voltage overhead transmission lines: Evidence from a causal approach in rural France

1. Introduction

The new energy economy around renewables will require massive investments in transmission infrastructure - particularly high voltage power lines to bring renewable (wind and solar) energy to market and places of consumption. In the United States, the “Building a Better Grid” initiative was recently launched to support the need for electricity transmission infrastructure (ETI) that is critical to achieving US goals of 100% clean electricity by 2035 and a zero-emissions economy by 2050.¹ In Europe, an investment of around €150 billion is planned for the installation of 48,000 km of new or upgraded transmission lines by 2030 (ENTSO-e, 2014). However, the installation of new electricity transmission infrastructure frequently meets with opposition from local host communities (Battaglini et al., 2012; Cohen et al., 2016). Such resistance generates costly delays and may interrupt or at least slow the transition to a low-carbon or carbon-free power system (Rious et al., 2011).

A familiar explanation for local opposition to infrastructure projects is the “Not In My Back Yard” (or NIMBY) syndrome. While people are globally in favour of the energy transition, they are reluctant to bear alone the external costs of energy facilities siting (Sun et al., 2023). Visual obstructions related to high-voltage overhead transmission lines (HVOTLs) have been identified as the main reason for people living close to such infrastructures speaking out against them (Cohen et al., 2014). Visible power lines, pylons, and transformers affect local residents’ perceptions of the loss of ecological and aesthetic values of the landscape (Soini et al. 2011; Cain and Nelson 2013; Lienert et al., 2018). Local residents are

¹ <https://www.energy.gov/articles/doe-launches-new-initiative-president-bidens-bipartisan-infrastructure-law-modernize>

also concerned about health risks from exposure to electromagnetic fields (Wadley et al., 2019).

Hedonic pricing models provide an empirical framework in which to make an economic valuation of these disamenities of energy infrastructure as reflected by variations in real-estate market prices (Gibbons, 2015; Maddison et al., 2022). As people choose their residential location according to their preferences and aversions (Tiebout, 1956), the hedonic price approach, formally developed by Rosen (1974), provides a theoretical and methodological framework for retrieving the implicit price of – or willingness-to-pay (WTP) for – environmental characteristics that do not have a specific market price. In principle, this empirical process is straightforward. In practice, there are a number of major modelling issues to be considered, such as the choice of appropriate instruments for measuring environmental amenities and the selection of an econometric specification by which to identify and interpret their impact on welfare (Bishop et al., 2020).

In the literature, one way to assess the possible impact of HVOTL infrastructure on house prices is to introduce a variable for the distance between individual houses and the transmission line pylons. Colwell (1990) was one of the first to show that the further away a house is from an ETI the higher the property's sale price. Subsequent studies measure visual nuisance more precisely using indicators that supplement distance to the infrastructure with the visibility of the line, and even more so of a pylon. Hamilton and Schwann (1995) mobilize the sale prices of individual houses (N=12,907) in Vancouver (Canada) and show that the value of a house located 100 m from a power line is 6.3% lower if the line or at least one tower is in the field of vision. For houses located more than 200 m away, the reduction in property values is around 1%. Des Rosiers (2002) uses information on 507 transactions in a southern suburb of Montreal (Canada) to suggest that it is the view of the pylons that is negatively correlated with property prices, with a premium ranging from -5% to -20%. He

suggests that the effect is highly local, varying from 50 m to 100 m from the pylon. Beyond 150 m, the effect appears to be insignificant. Sims and Dent (2005) adopt the same methodology to study 664 transactions near Glasgow (Scotland) and find that properties located within 100 m of power lines are associated with a 6% to 17.7% lower sale price or thereabouts. The actual view of a pylon is associated with a 7.1% to 14.4% lower sale price.

So far, most applications have been based on a correlational approach, which may differ from a causal analysis. A causal approach in hedonic pricing models usually relies on the application of quasi-experimental approaches to environmental valuation to address inefficiencies associated with externalities (Greenstone and Gayer, 2009). It is essential for transmission system operators to develop reliable estimates of the external costs of the presence of HVOTLs if they are to conduct informed negotiations about compensatory mechanisms with host communities. Using the conventional hedonic approach to capture the external cost of infrastructure might confound the estimate of HVOTLs' externalities with other non-observable neighbourhood effects that are correlated with those externalities (Kuminoff et al., 2010). Thus, the impact being measured may be correlational and not causal (Parmeter and Pope, 2013).

A recent empirical study by Tang and Gibbons (2021) analyses the effects of new HVOTL installations on home values for England and Wales, i.e. with more than 1.4 million transactions from 1995 to 2018. The analysis is based on a difference-in-differences approach where price changes around neighbourhoods that are close to overhead powerlines before and after the construction of new lines are compared to other transactions not involving any major visual changes. As compared to the previous literature, they assume that proximity to pylons and lines can affect the prices of properties up to 1200 m away. Their

101 analysis suggests that new overhead power lines may reduce prices by 4%
102 on average.

103 The present paper contributes to the literature on evaluating the impact of HVOTLs by
104 using a quasi-experiment that takes a different perspective. In making causal inferences about
105 the perception of the market, two conditions are important: (i) economic agents (buyers and
106 sellers) must be aware that a change will occur, and they must know where and when it will
107 occur; and (ii) it must be possible to confine the nuisance under study to some specific
108 distance. The analysis is based on single-family home transactions between 2003 and 2014
109 (construction start date) in the municipalities concerned by the “2-Loires” project to upgrade a
110 225 kV line linking Saint-Privat-d’Allier to Saint-Etienne in France. It is assumed that the
111 announcement of the upgrading plan, in 2009, might have affected public perception of a
112 forthcoming change in the nuisances caused by the HVOTLs in place. Importantly, the
113 announcement of the upgrading project did not mark a change in the footprint of the
114 infrastructure, thus expectations about the environmental benefits of removing the pylons and
115 undergrounding the power line are the only things that changed. Moreover, about 70% of the
116 buyers had their previous residence in one of the municipalities in the target area, and about
117 30% bought a property in the same municipality as their previous dwelling. It can be thought,
118 then, that the economic agents were sufficiently informed to be able to anticipate the impact
119 of the change.

120 The paper is original in three ways. First, the study takes advantage of the
121 announcement about the replacement of an existing transmission line, where it was decided
122 that part of it was to be undergrounded, to evaluate the impact of the prospect of the removal
123 of the HVOTL, i.e., before the transformation actually occurred. Working on the assumptions
124 of rational expectations and full information, the results of a difference-in-differences

comparison concerning the announcement of an undergrounding project could be interpreted as the anticipated net impact of HVOTLs on environmental amenities in the study area.

Second, assuming, as in most of the dedicated literature, that the impact is concentrated locally, the methodological framework adequately isolates for potential omission variable bias by limiting the selection of both treatment and control transactions within 1000 m of the existing pylons. This choice greatly reduces the sample size for the estimations, while controlling both for spatially unobservable characteristics and spatial autocorrelation among residuals.

Third, the estimation results are used to investigate a cost-benefit analysis using a Monte Carlo experiment approach. This exercise aims to limit the impact of a large confidence interval when investigating the profitability of a public project. It indicates a probability of profitability regarding the estimation of benefits and costs instead of relying on a single point estimation.

The results show that the negative impacts of HVOTLs may result in a price depreciation of about 40% for houses located within a strip of 200 m of the existing pylons. While the number of observations for comparison is small, a series of robustness checks suggests that the conclusions are consistent as regards the choice of control transactions. The results of the difference-in-differences approach indicate that the net impact of HVOTLs on house values may be much higher than previously reported in the hedonic literature. Furthermore, a benefit–cost ratio analysis for the undergrounding project, similar to stated-preference estimates (Navrud et al., 2008; Tempesta et al., 2013), is conducted to investigate the profitability of such actions. From a public policy perspective, the results suggest that opting for underground transmission lines may be a valuable tool, if not necessarily a panacea, for optimizing social welfare as well as global economic benefit as a Monte Carlo experiment suggests that the ratio is positive in about 85% of cases.

The paper consists of six sections in all. Section 2 sets out the methodological framework. Section 3 outlines the data used for the analysis. Section 4 shows the results and sensitivity tests. Section 5 presents a benefit-cost analysis. The final section concludes.

2. Methodological Framework

According to the hedonic pricing model (Rosen, 1974), the sale price of a complex good i at time t , stacked in the vector $\mathbf{y}_{it}$, can be expressed as a function of all its characteristics, including individual and spatial ones, synthesized in the matrix $\mathbf{X}_{it}$. The coefficients related to the independent variables, $\boldsymbol{\beta}$, can be used to retrieve the implicit (or hedonic) prices of each individual characteristic (equation 1).

$$\ln(\mathbf{y}_{it}) = \mathbf{1}\alpha + \mathbf{D}_{it}\boldsymbol{\delta} + \mathbf{X}_{it}\boldsymbol{\beta} + \boldsymbol{\epsilon}_{it} \quad (1)$$

The hedonic pricing model is usually based on a logarithmic transformation of the dependent variable, $\ln(\mathbf{y}_{it})$, to better control for heterogeneity. The vector of the dependent variable is of dimension $(N_{\tau} \times 1)$, where N_{τ} is total number of observations collected over time period t ($\tau = 1, \dots, T$).² The matrix of independent variables, $\mathbf{X}_{it}$ is a matrix of dimension $(N_{\tau} \times K)$, $\mathbf{D}_{it}$ is a matrix of time (year) fixed effect variables of dimension $(N_{\tau} \times (T-1))$, $\mathbf{1}$ is a vector of dimension $(N_{\tau} \times 1)$ of elements equal to 1, and $\boldsymbol{\epsilon}_{it}$ is an error term, assumed to be independent and identically distributed of dimension $(N_{\tau} \times 1)$. The vector of parameters $\boldsymbol{\delta}$, of dimension $((T-1) \times 1)$, captures the nominal change in the house price over time, the vector of parameters $\boldsymbol{\beta}$, of dimension $(K \times 1)$, is the vector of implicit prices of the individual characteristics, while the scalar parameter α is the constant term.

While it is a straightforward matter to estimate the equation price model, causal inference cannot be estimated in such a cross-sectional specification. To make such

² The total number of observations is given by $N_{\tau} = \sum_t N_{it}$.

estimations, standard methods have been developed (Antanokis et al., 2010), including the difference-in-differences (DID) approach. The DID approach is a simple extension of the hedonic pricing model where two additional variables are accounted for.

The first variable serves to isolate the transactions recorded before or after a specific (exogenous) change ($DT_{it} = \{0,1\}$). The variable takes a value of zero if the transaction is recorded before a certain date ($t < t^*$), while it takes a value of one if the transaction is observed on or after that date ($t \geq t^*$).

The second variable expresses the relative location of the transactions ($DD_{it} = \{0,1\}$). The variable takes a value of one if a given house is located within a specific zone liable to be impacted by the change (treatment area). This zone is located within a specific distance, $d_{ip} \leq d_c$, with d_{ip} being the distance between a transaction i and a pylon p , and d_c being a critical distance cut-off value identifying the within/outside (treatment) zone. It takes a value of 0 if it is located further away ($d_{ip} > d_c$).

Introducing both variables, as well as the cross-product of the variables, into the original price model equation allows us to obtain the DID specification (equation 2). This equation enables us to retrieve the causal effect of the change under study when the treatment and control groups are adequately identified, with $\mathbf{DD}_{it}$ a vector of dimension $(N \times 1)$, $\mathbf{DT}_{it}$ another vector of dimension $(N \times 1)$, and $\circ$ the Hadamard (term-by-term) matrix product.

$$\ln(\mathbf{y}_{it}) = \mathbf{1}\alpha_0 + \mathbf{D}_{it}\delta + \mathbf{X}_{it}\boldsymbol{\beta} + \mathbf{DT}_{it}\eta + (\mathbf{DD}_{it} \circ \mathbf{DT}_{it})\theta + \boldsymbol{\varepsilon}_{it} \quad (2)$$

The scalar parameter δ controls for any differences there might be between the transactions within each year, the scalar η controls for differences that might occur for both groups after the change, while the scalar θ controls for the difference that occurs for the treatment group after the change occurs. It is the parameter θ that allows us to isolate the effect of the treatment on house prices.

195

196 **3. Data**

197 **3.1. Description of the project under study**

198 The study concentrates on the project to replace an existing transmission line between
199 Saint-Privat-d'Allier and Saint-Etienne in the Auvergne-Rhône-Alpes region of France. The
200 original 25 kV line was built in 1941 and was in need of major investment to reduce the risk
201 of shut down. To avoid such a situation, it was decided, on 19 March 2009, to rebuild the line.
202 It was decided to replace the entire length of the power line. The construction was split into
203 three separate phases.

204 The official announcement of the technical solution, which included undergrounding a
205 part of the line, was set for February 2010. At this time, *Réseau de transport d'électricité*
206 (Rte), the public network operator responsible for electricity transmission in France, opted to
207 reconstruct the line as a double (instead of a single) circuit. The new transmission line was to
208 feature about 250 pylons and three underground sections. It was to cross 23 municipalities
209 (*communes*) for a total investment of about €132.7M under the conditions prevailing in 2012.
210 Of this total investment, about €21M was earmarked for compensating, on a voluntary basis,
211 local authorities for expected damage.

212 The route was finalized during the period 2012–2013, and the final public
213 announcement was made in June 2014. While the details about the exact location of the new
214 pylons were not entirely determined at that time, it had already been decided that a short
215 section of the line would be undergrounded to bypass the main town along the line. The total
216 length of the underground line was about 8 km out of a total 87 km. The replacement work
217 began officially in 2015 and the line was officially brought into service on 1 December 2017,
218 although the old line was only fully dismantled in March 2018 (see Figure 1).

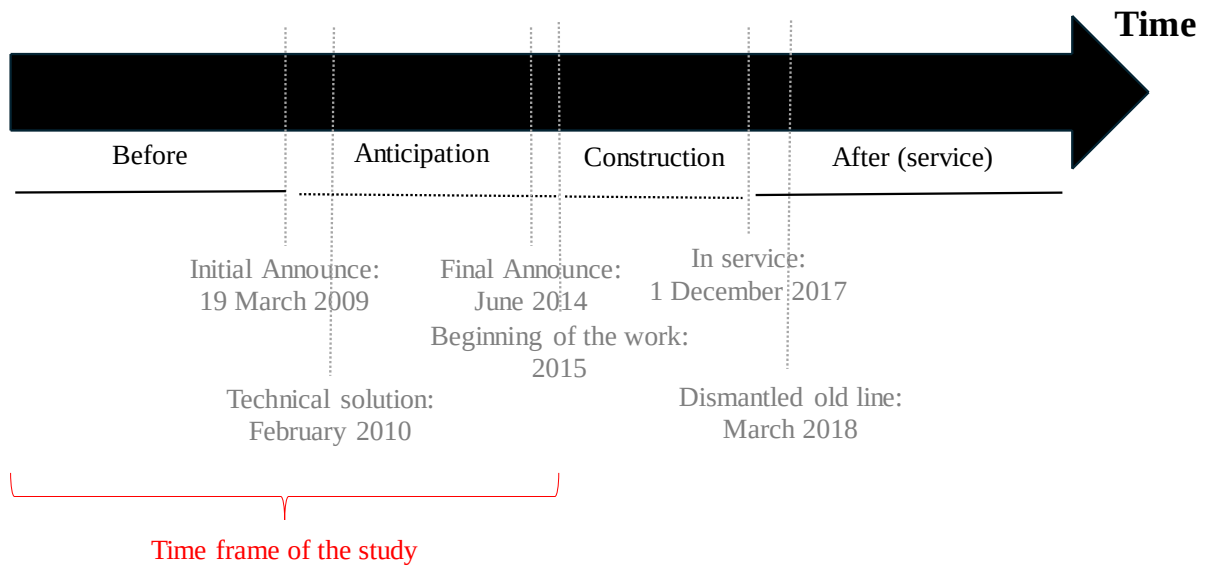


Figure 1: Time frame of the full project and the case study

The analysis focuses on the period during which the public consultation about the upgrading project was carried out, i.e., right after the announcement regarding the decision to rebuild the transmission line, and just before actual construction began. This period was essentially given over to delineating the route of the new high voltage transmission line. Studying this period means we can evaluate the impact of the prospective change, as opposed to the actual change. The paper concentrates on the shift in market expectation, as data on transactions after the project was completed are unavailable.

For the remainder of the paper, it is assumed that the decision to rebuild the power transmission line resulted, from the real-estate market perspective, in an exogenous shock, especially the decision to include the underground segments. This is so because the public network plan (2010–2020) to rebuild the line did not become public information until the official announcement was made. Moreover, while discussions were going on about the new route for the line, RTE had already decided on the location of the underground segments. This provides a quasi-experimental design to focus on the perception of the negative impact of the

235 line by testing whether the market internalizes this major change through higher sale prices
236 after the announcement.

237 The anticipation effect related to announcements about projects is familiar enough
238 when it comes to implementing and developing new transport infrastructure (Devaux et al.,
239 2017; Murray and Bardaka, 2021) but also energy infrastructure (Boslett et al., 2016). Thus,
240 the change in the perception of proximity to/view of such infrastructure is based on a real
241 situation, and not on a survey, where the agent does not necessarily feel that the change is for
242 real. Thus, anticipation is measured from a different perspective based on actual transaction
243 data where it is already known that a public intervention is to take place. The change thus
244 measures perception from the perspective of actual experience.

245 **3.2. Transaction data**

246 The information about single-family home transactions was provided by the *Notaires*
247 *de France* information system (also called *Perval*). The full set of transactions covers the
248 period between January 2000 and December 2016 for a total of 3583 single-family house
249 sales for the whole region under study (Figure 2).

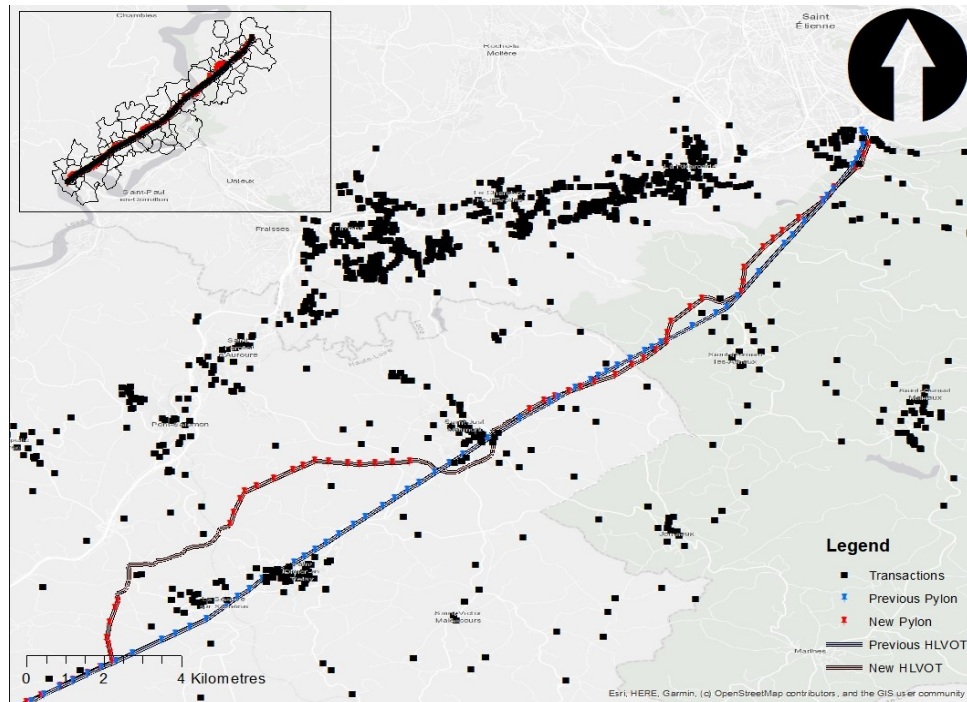


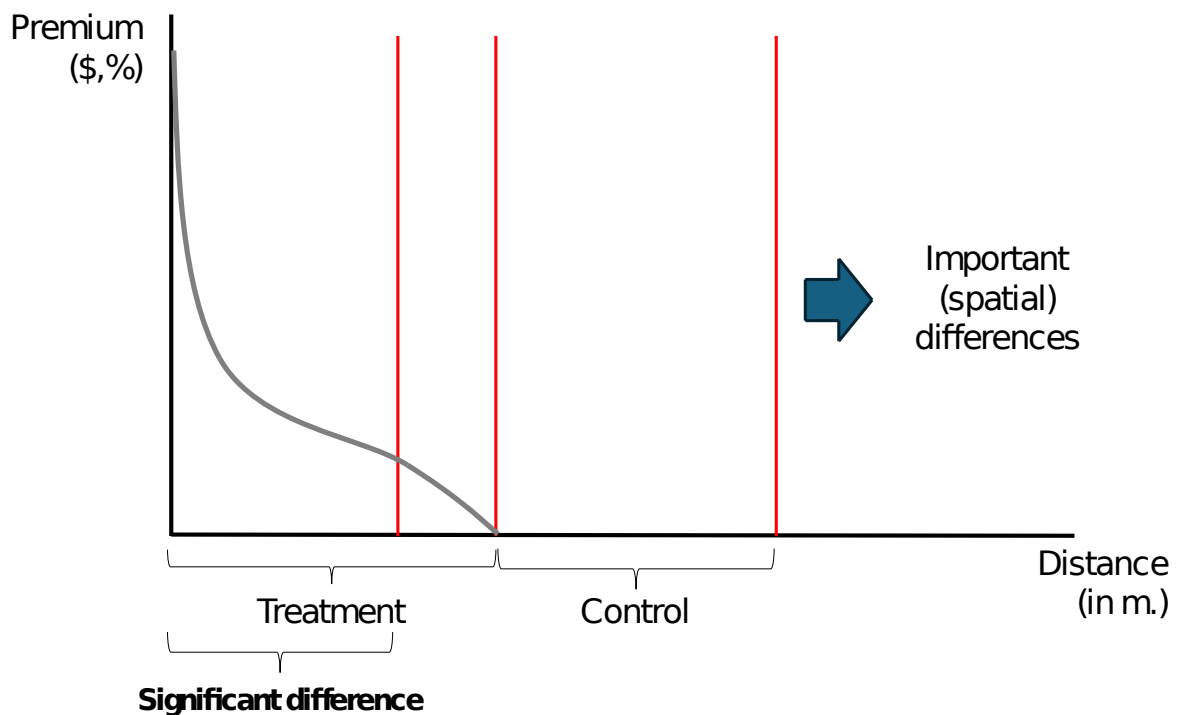
Figure 2: Spatial distribution of transactions in the region under study

While the reconstruction of the transmission line may affect many municipalities, the literature recognizes that the nuisance impact, of transmission lines is closely concentrated around the pylons. In the case of the “2-Loires” project, it was decided to concentrate most of the reconstruction within 50 m of the existing pylons. Usually, the presence of a power line commands a restriction on land use within 100 m on either side and the new line itself is some 20 m wide. Some studies suggest that the negative effect is closely related to the encumbrance of the pylon (Des Rosiers, 2002) and is usually inexistant beyond 150 m, but there remained some uncertainty regarding the exact location of the pylons along the new line. For those reasons the treatment area was first defined to fit a 200 m distance to the existing pylons (and not the distance to the line).³

To adequately isolate the local environmental disamenities on house price, the definition of the treatment and control area must be spatially concentrated. As the impact

³ As the pylons are relatively close together, the distance to the line is highly correlated with the distance to the nearest pylon. For this reason, the analysis only focuses on the distance to pylon.

265 might decrease from the distance to where change occurs, and that the impact might be non-
 266 significant after a certain distance, leave a zone outside of the analysis should help to identify
 267 adequately treatment effect (Figure 3). Otherwise, the analysis might suggest a false non-
 268 significant impact if the number of observations within the non-significant area is important
 269 and drive the mean. For this reason, the treatment zone might be smaller than the control area.
 270 However, the control area might also be limited to make sure that the comparison is based on
 271 places where spatial unobserved characteristics are similar, limiting potential omission
 272 variable bias problem.



273
 274 Figure 3: Potential spatial distribution of (positive) change in environmental amenities

275 Moreover, to ensure that the perception effect is related to a major change, the analysis
 276 focuses explicitly on the area in which it is proposed to underground the new line. For this
 277 part of the project, and to avoid potential problems of violation of the stable unit treatment
 278 values assumption (SUTVA),⁴ the control group is defined by a zone twice as wide as the
 279 treatment group using transactions located within 400 m and 800 m. This leaves a 200 m

31 ⁴ Which is related to the spatial spillover effect.

distance between the treatment and control groups (Figure 4). This local approach is also designed to control for high heterogeneity in the spatial amenities that might otherwise bias the analysis.

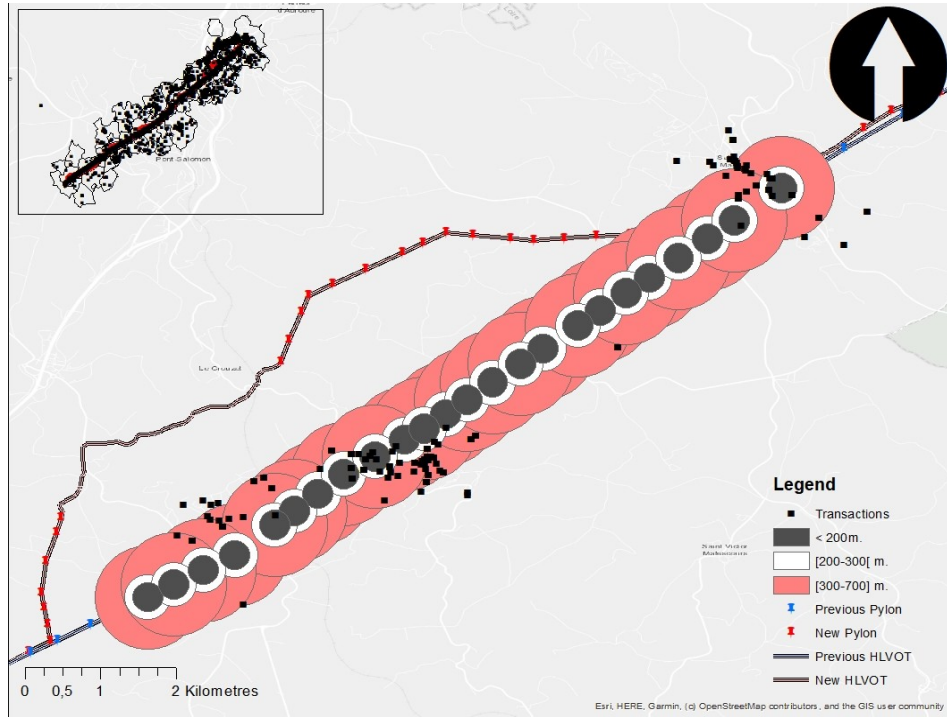


Figure 4: Identification of the transactions under study

Filters are applied to the data set. First, as the study focuses on the anticipation (and adjustment) effect related to the announcement of the modification, transactions in 2015 and 2016 are discarded.⁵ Second, a filter on the final sale price and the lot size is used to remove the extreme values that might influence the estimation results, especially when limiting the analysis to a small sample size. Finally, to make sure that the number of transactions before and after the announcement is not highly unbalanced, it is proposed to keep transactions recorded between 2003 and 2014 in the analysis. In the end, the total sample size available for the analysis is fixed at 117 transactions (Figure 2).

The number of transactions within the treatment area is small. A total of 23 transactions was observed in the period 2003–2014. Of this total, 16 transactions were

⁵ It should be noted that only three transactions is available for 2015, while 8 transactions is available for 2016.

observed before the announcement and 7 transactions after within 200m from the pylons.⁶ The number of transactions within the control area is greater (94 transactions), but a similar pattern is observed with more transactions before the announcement (56) than after (38)(Table 1).

Table 1: Identification of the treatment/control groups by zone

	Distance to the old pylons			
	< 200 m.		]400; 800] m.	
Time	N. obs.	Sale price (in €)	N. obs.	Sale price (in €)
Before March 19 2009	16	164965.6	56	129799.4
After March 19 2009	7	182428.6	38	138158.2
Between 2003 and 2014	23	170280.4	94	133178.5
Note: All prices are in nominal values				

A quick look suggests that the mean sale price rose more in the treatment area (an increase of about €17,463) than in the control area (increase of about €8,359). However, this comparison hides the fact that the values are in nominal terms, as well as the fact that the characteristics of the houses may be different.

Focusing on the descriptive statistics of the transactions, the characteristics of the transactions used for the analysis ($N_{\tau} = 117$) are not statistically different from the characteristics of the transactions for the period 2000–2014 ($N_{\tau} = 146$). The only statistical difference is in the sale price and the sale year (Table 2). However, these differences can easily be explained by the larger span of the time period. The mean sale price is about €140,472 for a mean land lot size of about 804 m². The houses have about 4 rooms, 1 bathroom, a parking space, and most of the transactions involve detached single-family houses.

⁶ Adding the information from the construction period (2015–2016) has only a slight influence on the size of the groups and on the results. It adds three observations to the treatment group after the announcement (all in 2016). To be consistent, the analysis is based on the announcement period only so as to avoid any undesirable cross-effects between specific periods. However, analysis with the transactions confirms the estimation results.

On the one hand, limiting the analysis to a specific area obviously limits the sample size. However, the investigation is spatially concentrated on transactions that will undergo a marked change, i.e. the dismantling of the existing pylons and the undergrounding of the transmission line. Thus, it is in this area that any sizeable influence on sale prices might occur. Moreover, the selection of the sample does not influence the mean statistics of the selected transactions (Table 2).

On the other hand, limiting the analysis to municipalities (*communes*) affected by the undergrounding of the transmission lines helps limit the possible omission bias related to spatial characteristics. The municipalities affected have about the same population size and the spatial characteristics and amenities are similar within the small municipalities.⁷ The inclusion of a set of municipalities dummy variables aims to control for specific spatial differences within the municipalities. Limiting the extension of the spatial area covered also limits possible problems related to spatial autocorrelation among residuals.

Two conditions must be met for changes in home sale prices to reflect environmental disamenities. The first is that potential buyers (and sellers) should be aware of the existence of infrastructure in the vicinity of their future purchase and the dismantling of the infrastructure and therefore take this information into consideration when making an offer for a house. This seems to be the case: more than 70% of them already had their previous residence in one of the municipalities of the area targeted by the “2-Loires” project.

The second condition is that the market should perceive the proximity of the infrastructure as a nuisance. About 30% of homebuyers bought a property in the same municipality as their previous dwelling. In other words, we can comfortably assume that, before buying the house, the great majority of purchasers knew about the presence of the HVOTL near their future residence.

⁷ The number of dwellings was about 2032 in 2018 in Saint-Just-Malmont (the closest town to St-Etienne), while it was 1762 for Saint-Didier-en-Velay and 794 in La-Séauve-sur-Sermène.

337 | Table 2: Descriptive statistics of the sample
338

Variables	Full sample			
	Mean	s.d	Min	Max
Sale price (nominal €)	121759.60	64809.46	22867	390000
Sale price (log)	11.55	0.599	10.04	12.87
Distance to closest pylon (new) (m.)	1721.81	348.489	1040.73	2195.28
Distance to closest pylon (old) (m.)	381.80	151.179	38.47	695.70
Lot size (m ²)	869.04	990.918	35.00	7864.00
Lot size (m ²) - log	6.21	1.146	3.56	8.97
Number of rooms	3.92	1.934	0	6
Number of bathrooms	0.96	0.513	0	2
Number of parking spaces	0.88	0.519	0	2
Sale year	2006	3.904	2000	2014
Number of floors	1.70	0.883	0	3
Townhouse	0.34	0.474	0	1
Detached single-family houses	0.52	0.501	0	1
Distance to St-Etienne (km)	16.55	2.130	12.36	19.93
Variables	Selected sample			
	Mean	s.d	Min	Max
Sale price (nominal €)	132405.80	64044.45	22867	390000
Sale price (log)	11.66	0.561	10.04	12.87
Distance to closest pylon (new) (m.)	1679.07	352.413	1040.73	2195.28
Distance to closest pylon (old) (m.)	373.85	152.988	38.47	695.70
Lot size (m ²)	886.76	963.962	35.00	7864.00
Lot size (m ²) - log	6.26	1.141	3.56	8.97
Number of rooms	3.99	1.899	0	6
Number of bathrooms	0.99	0.516	0	2
Number of parking spaces	0.90	0.536	0	2
Sale year	2007	3.322	2003	2014
Number of floors	1.68	0.921	0	3
Townhouse	0.34	0.474	0	1
Detached single-family houses	0.54	0.501	0	1
Distance to St-Etienne (km)	16.33	2.256	12.36	19.49
Note: Full sample size is 152 transactions; Selected sample size is 125 transactions				

339

340 4. Results

341 4.1. Estimation results

Before turning to the estimation, a brief remark is called for about the small sample size. First, since the number of observations in the treatment area (within 200 m of the existing pylons) is small, it is impossible to formally test for the parallel trend assumption. However, as the transactions are spatially highly concentrated, the assumption appears to be quite realistic, as the price within a distance of 1,600 m (radius of 800 m) within a rural community is likely to share the same trend. Second, since no transactions occur between January and March 2009, the vector $\mathbf{D}_{it}$ is perfectly correlated with the vector $\mathbf{DD}_{it}$ (see equation 2). For this reason, one of the terms in the original equation is deleted in the estimation process.

The models are estimated using *Stata* software. All models are corrected for a robust variance-covariance matrix.⁸ The basic model includes, on the one hand, a set of individual characteristics: i) lot size (in log transformation); ii) number of rooms; iii) presence of a basement; and iv) period of construction. It also includes, on the other hand, a set of independent variables controlling for: i) spatial fixed effects at the municipality level; and ii) temporal fixed effect for year of transaction, controlling for nominal evolution of price.

The estimation results suggest that the basic hedonic price equation is able to explain more than 63% of the variance in the sale price (in log). The model is globally significant, while the coefficients related to the individual and spatial characteristics are consistent with theoretical expectations: houses with better amenities sell for higher prices (Table 3 – column 1). Spatial autocorrelation is detected among the residuals of the models, as suggested by the slightly significant Moran's I index, which suggest that spatial component in house price determination is not adequately taken into account in this basic specification.

⁸ With the robust option.

364 | Table 3: Estimation results and robustness checks

Variables	Base Model		DID model		Robustness (1)†		Robustness (2)‡	
	Coefficient	sign.	Coefficient	sign.	Coefficient	sign.	Coefficient	sign.
Lot size (m ²) - log	0.2844	***	0.2835	***	0.2754	***	0.2294	***
Number of rooms (ref: 2 or less)								
3	-0.3285		-0.3327		-0.2829		-0.0884	
4	-0.2669		-0.2564		-0.2104		-0.0327	
5	-0.0711		-0.0675		-0.0326		0.1381	
> 6	-0.1726		-0.1239		-0.0969		0.0791	
Number of bathrooms (ref: 0)								
1	0.0269		-0.0038		-0.0607		-0.1028	
2	0.2517		0.2285		0.2006		0.0890	
Number of parking spaces (ref: 0)								
1	0.0990		0.1200		0.0684		0.1469	
2	0.1773		0.1717		0.1303		0.2062	
Sale year (ref: 2003)								
2004	0.3642	*	0.3532	*	0.3460	*	0.1894	
2005	0.4003	**	0.4019	**	0.4343	**	0.3299	**
2006	0.5663	**	0.5739	**	0.5801	*	0.4257	*
2007	0.4517	***	0.4511	***	0.4404	***	0.3684	**
2008	0.3302		0.3307		0.3278		0.3053	*
2009	0.5090	*	0.3175		0.3221		0.3492	
2010	0.3159		0.3352		0.1078		0.3340	
2011	0.8141	***	0.6941	***	0.6666	***	0.5376	***
2012	0.7532	***	0.7357	***	0.7309	***	0.6194	***
2013	0.3966	*	0.3313	*	0.3093	*	0.2163	
2014	0.7351	***	0.6850	***	0.6629	***	0.4885	***
Number of floors (ref: 0)								
1	0.0584		0.0087		0.0303		-0.1069	
2	-0.0345		-0.0807		-0.0497		-0.1664	
3	-0.0002		-0.0218		-0.0277		-0.1026	
Townhouse	0.1932		0.2255		0.1490		0.1601	
Detached single-family house	0.3691	*	0.3824	**	0.3356	*	0.3480	**
Distance to St-Etienne (km)	-0.0583	***	-0.0571	***	-0.0536	**	-0.0458	***
Within 200 m of old pylons			0.0460		0.0308		0.0965	
Within 200 m after announcement			0.3360	*	0.3323	*	0.3315	*
Constant	10.1559	***	10.1480	***	10.2444	***	10.3784	***
N.obs.	125		125		118		162	
F-Stat	13.93	***	14.43	***	13.56	***	11.17	***
R ²	0.6479		0.6655		0.6536		0.6293	
RMSE	0.37471		0.3690		0.3713		0.3543	
Moran's I Index	-0.0126		-0.0157		-0.0172		-0.0126	
Legend: *** p < 0.001; ** p < 0.01; * p < 0.05								
† Analysis is made using transactions within 300–600 m of the old pylon as the control area								
‡ Analysis is made using transactions within 300–800 m of the old pylon as the control area								

365

A few specific points are called for about the interpretation of the results for the base model. First, it appears that lot size is one of the most important factors in explaining the final sale price. Lot size appears to be the characteristic that accounts for most of the heterogeneity in the goods in such areas. . Second, house price is higher for properties having 5 or 6 rooms and sale price is higher for more recent properties. Third, house price vary according to their relative location, with prices being relatively lower in two municipalities that are further away from the closest major city, St-Etienne. Fourth, the year dummy variables are globally significant, suggesting a rise in nominal price over time.

Focusing on the second model with the original DID specification (Table 3 – column 2), the results suggest that the announcement of the removal of the pylons and the undergrounding of the transmission lines positively and significantly affected sale price. While the price was about the same before the announcement after controlling for the characteristics of the houses, as suggested by the non-significant coefficients related to proximity to the old pylons, the price premium related to the announcement of the project was about 38%. The positive price premium for the houses located close to pylons before the announcement can be explained by the fact that those houses are located in the centres of the small municipalities.

Based on the mean sale price of houses located close to the pylons before the announcement, i.e. about €164,965, the price premium represents a rise of about €62,958 in nominal value, which is considerable. Of course, the size of sample is small, which results in large standard errors. The confidence interval of the estimated parameter suggests that the price premium lies between €6,176 and €138,580. Thus, while the numbers seem high, we might be cautious about extrapolating the estimated price premiums according to the sample size. Nevertheless, the main conclusion is that the announcement of the removal of the pylons was positively perceived by the real-estate market.

4.2. Robustness checks

To check the robustness of the results, three complementary analyses were conducted. The first one is based on changing the definition of the treatment area. The second one is based on changing the definition of the control area. The third one is based on a non-parametric approach to test the significance of the original estimation using a randomization test approach (Dubé et al., 2023) on the original choice of the treatment and control area.

The first robustness check analysis used the same time period (2003–2014) but confined different size for the treatment area. The treatment zone varies between 100m to 250m. The analysis suggests a decreasing impact with the distance to the nearest pylon (Table 4), however, the impact appears to be significant only between 150m and 200m only. On the one hand, the sample size is probably too small to return a satisfactory statistical performance for a treatment area of 0-100m. On the other hand, it also suggest that opting for a too wide treatment area might slightly reduce the estimate of the impact and affect the significance of the coefficient, which is consistent with theoretical expectation (see Figure 2).

The second robustness check analysis took the same time period but change the size of the control area. The control area starts from 400m to the nearest pylon, while the ending zone vary between 750m and 900m (Table 5). The first analysis suggests that taking a too small control area (400-750m.) return a non-significant impact. This result might be related to the small sample size within the control area, and the number of observations used to estimate the numerous numbers of parameters.⁹ For the remaining analysis, the impact of the announcement suggests a robust, positive and significant price premium. The significant price premium eliminates the residual spatial autocorrelation detected using the 400-750m control area. In both specifications, the treatment effect estimated remains similar and about 38%.

⁹ The same analysis has been made using a treatment area of 0-150m. and the results are qualitatively similar.

--	--	--	--

[illegible]

415

416 The final robustness check analysis is based on a permutation test to calculate the significance

417 of the parameter of interest in the original specification. A non-parametric test is used to

418 calculate the pseudo p-value for the parameter of interest. The idea is to permute the variable

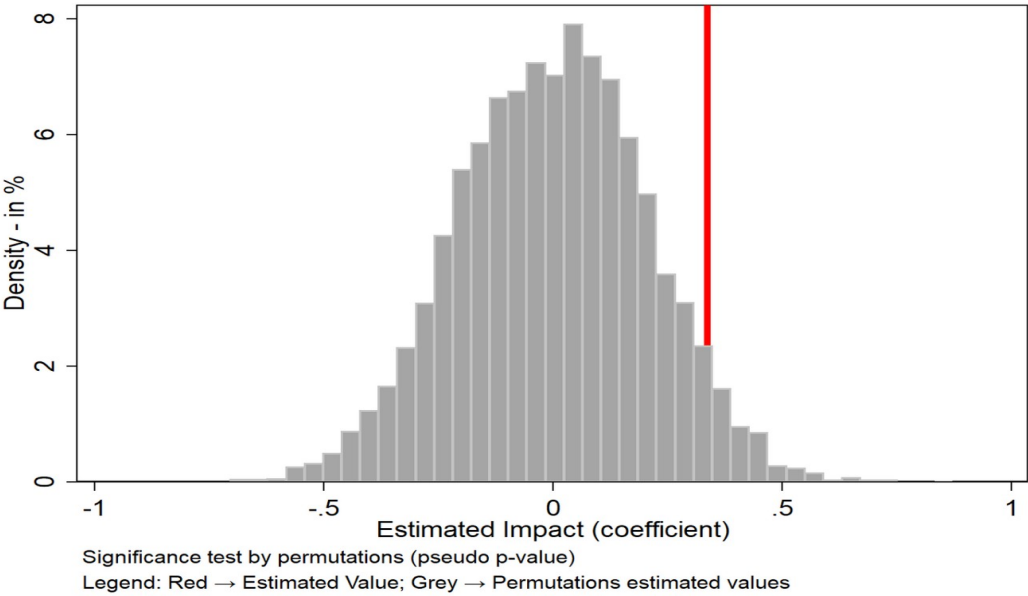
419 of interest (here the treatment variable) randomly for the before and after time periods and to

re-estimate the model. The original parameter is significant if it is distinct from the one

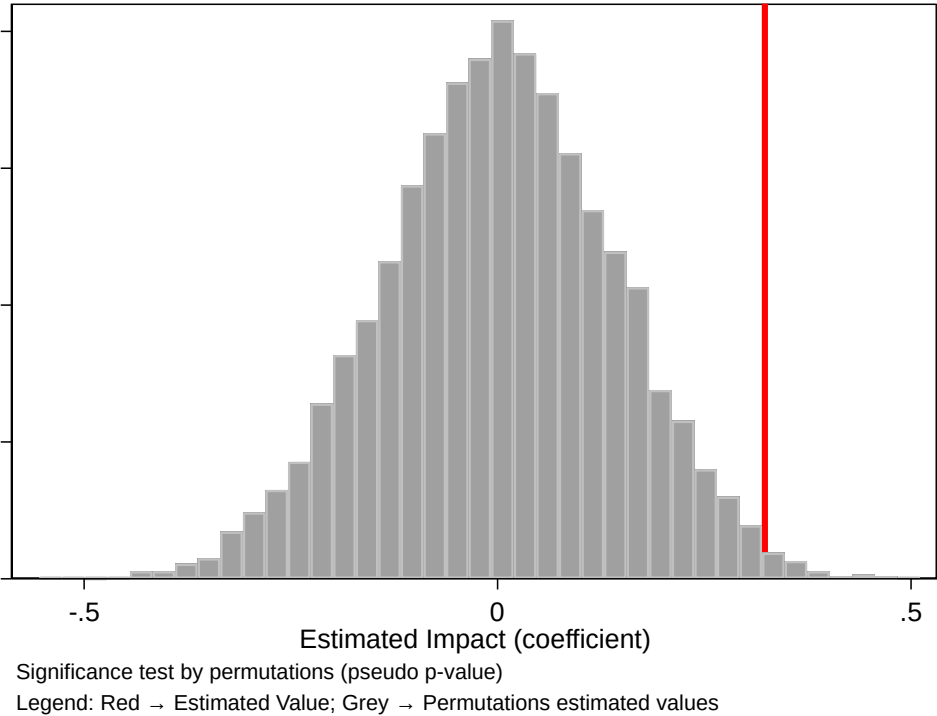
421 estimated using the permutation. Otherwise, the model is not able to give a clear difference

422 between the estimated parameter on the real data set and what can be obtained with a pure

423 random distribution. Stated otherwise, the original parameter must be far enough from the
424 distribution of the parameters based on their fictive status to be significant. The test is based
425 on 9999 permutations and the rank of the original parameter is calculated to obtain the pseudo
426 p-value. The results suggest that the parameter is statistically significant, with a pseudo p-
427 value of 0.0092 (Figure 5). In the end, the statistical conclusion appears to provide a robust
428 impact coefficient, even if the sample size used to make the analysis is small.



429



430

Figure 5: Determination of the pseudo p-value using a permutation test.

5. Evaluation of the (partial) economic impact of the undergrounding project

One way to avoid the negative impact of high voltage transmission lines is to opt for a fully underground line. In such a case, the negative nuisance of the line is almost non-existent (Navrud et al., 2008; Menges and Beyer, 2014; Lienert et al., 2018). While this solution appears to be suitable for avoiding many downsides, it is sometimes deemed too expensive. For example, a project initiated in 2003 for a new transmission line between Spain and France through the Pyrenees Mountains was finally turned into an underground line after discussions with the local population. The original project for an overhead transmission line aroused strong opposition and it took about eight years of consultation to finally propose the underground solution (Ciupuliga and Cuppen, 2013). The line was put into service in 2015 and the final cost of the 33 km line was €357M, including an additional cost of €257M related to the choice of undergrounding.¹⁰

As the statistical analysis suggests, there is an individual and economic cost to living close to an overhead transmission line, at least for people directly exposed to the infrastructure. The availability of estimates of the causal effect of the presence of overhead transmission lines can be used to develop a partial cost-benefit approach to undergrounding schemes based on the resident household's willingness-to-pay to have an energy infrastructure-free landscape.

On the one hand, the marginal WTP, evaluated between €14,079 and €148,595 from the analysis, can be used to estimate the environmental net benefit values, as the sample is representative of the entire geographical area (Bateman et al., 2006). A total of 650 single-

¹⁰ Which represents a rough estimation of about €7.79M/km.

family residences, generally owner-occupied, were located within 200 m of the source of the nuisance, which amounted to about 2000 inhabitants. On the other hand, the cost per km of an underground high voltage line varies from project to project. In the case of the “2-Loires” project, the cost is estimated at between €1M and €2.8M. Based on those intervals, it is possible to evaluate the net benefit of undergrounding the transmission line.

A first and simple analysis consists of taking the mean value of the estimation of the benefit and the cost to calculate the (partial) net benefit. In such a case, the benefit clearly outweighs the construction cost. The net benefit is about €31.58M. It would have been equal to zero if the total number of houses located within 200 m of the pylons had been 211, representing about 1/3 of the total housing stock. However, this estimation relies on a point estimate only and might suffer from the lack of observations related to the treatment group after the change occurs.

Another way to make the analysis is to use the extreme values of the intervals for the estimated benefit and the building cost of the different scenarios. On the one hand, the results suggest a total benefit that varies between €4M and €90M. On the other hand, the actual cost of building the infrastructure, for the 8 km stretch, suggests that the cost varied between €22.4M and €8.0M. In the worst-case scenario, i.e. when the estimated benefit is about €6.2K/house and the construction cost is €2.8M/km, the net benefit is estimated to be about -€18.4M (Table 5). In such a case, the minimum number of houses affected by the presence of the pylons should be 3,627 to reach a zero net benefit. In the best-case scenario, i.e. when the estimated benefit is about €138.6K/house and the construction cost is €1.0M/km, the net benefit is estimated to be about €82.1M. In this case, the presence of only 58 houses located close to the existing pylons is necessary to ensure a zero net benefit (Table 5).

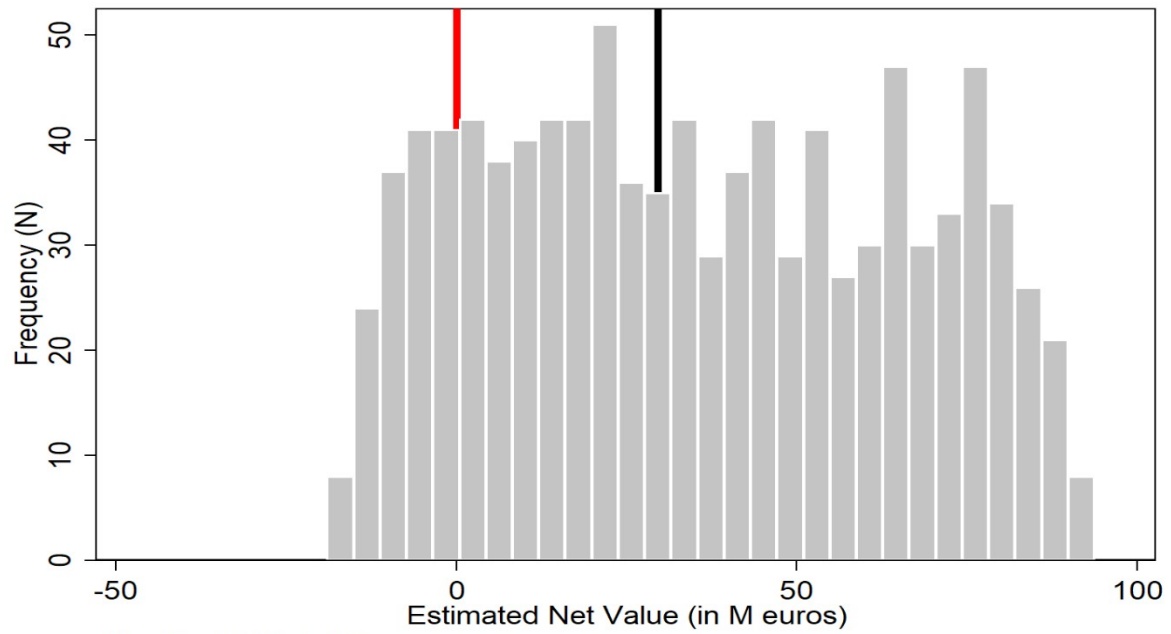
Table 6: Scenarios for benefit-cost analysis

Cost-benefit Analysis component	Negative	Mean	Positive
---------------------------------	----------	------	----------

Benefits (K€)			
<i>Impact/house</i>	3.66	68.91	158,33
<i># of house</i>	650	650	650
Estimation	2379	44793	102913
Costs (K€)			
<i>Cost/km</i>	2800	1900	1000
<i># of km</i>	8	8	8
Estimation	22400	15200	8000
Difference (M€)	-20.02	29.59	94.91
# of houses needed‡	6119	221	51
Note: ‡ to have a difference of 0			

479

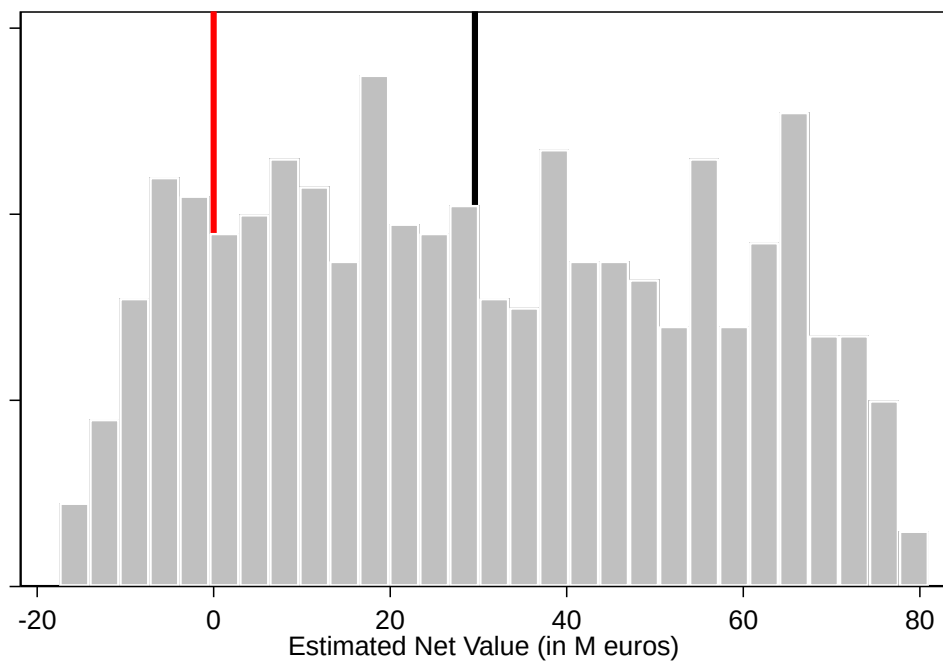
480 Another way to deal with the analysis is to conduct a Monte Carlo experiment using
481 the confidence interval regarding the benefits and the costs. Using 1000 simulations by
482 randomly changing the value of the benefit and cost, the (partial) cost-benefit analysis reveals
483 the distribution of the net benefit is mainly positive (Figure 6). The analysis suggests that less
484 than about 15% of the net benefit is negative, which in turn suggests that the decision to bury
485 the transmission line results in a positive impact for about 85% in the different scenarios. In
486 the end, the analysis returns similar results to those reported by two earlier stated-
487 preference studies (Navrud et al., 2008; Tempesta et al., 2013) conducted in Norway and
488 Italy. Even for sparse rural areas, it appears that the benefits from burying high
489 voltage power lines do exceed the costs. Of course, the analysis is only
490 partial since more benefits and costs could be included in a more complete
491 scheme. However, the benefit/cost ratio yields some interesting insights
492 for developing public policies about the future development of
493 transmission lines.



Result for 1,000 simulations

Legend: Red line → $\Delta = 0$; Black line → Point estimate

494



Result for 1,000 simulations

Legend: Red line → $\Delta = 0$; Black line → Point estimate

495

496

Figure 6: Support for positive benefit-cost ratio

497

498 6. Conclusion

499

500

This study contributes to evaluation of the impact of electricity transmission infrastructures on housing prices, as they produce disamenities (mainly visual effects, and

74

75

potential electromagnetic field emissions) for residents living close by. The analysis is based on the effects associated with the announcement of a project to upgrade a high voltage (225 kV) line in a rural area around the town of Saint-Etienne in France. More specifically, the announcement of the burial of a portion of the line (8 km) serves as a quasi-natural experiment with which to investigate the perception of the electric transmission pylons on house prices. As the analysis is based only on the impact after the announcement, and not on the final market adjustment, the paper aims at singling out the market reaction to a political decision of the kind.

To do this, the statistical analysis draws on difference-in-differences (DiD) estimators using a hedonic pricing model for three municipalities affected by the decision to switch from overhead to underground high voltage transmission lines. The selection of single-family home transactions located within 200 m of the pylons of the old line as the treatment group enables us to compare a situation with and without the presence of the infrastructures, for the same localities. Such an approach avoids confusion between the impact of the infrastructure and the effect of other environmental variables.

The results suggest that there is a negative and significant impact related to the presence of pylons on single-family house prices. The results show that the announcement of the intention to replace the transmission line is reflected by a rise of about 40% in the price of a house located within 200 m of existing pylons. For the area under study, this impact represents a price premium of about €68,912 in nominal value.

The large disparities in properties on the real-estate market in rural areas mean that the price premium can also vary quite considerably, from €6,176 to €138,580. Based on these estimations and on the estimation of the additional cost of undergrounding a power line (between €1M and €2.8M/km), a cost-benefit analysis is conducted. To investigate the robustness of the analysis, a Monte-Carlo simulation based on different combinations of the

cost and benefit is developed to build a distribution of the potential net benefits. The analysis suggests that about 85% of the point estimates return a positive net benefit, emphasizing the economic potential related to the undergrounding of transmission lines.

The paper is original in three ways. First, it uses a quasi-experimental approach based on an exogenous announcement about undergrounding a high voltage transmission line to investigate the causal effect related to such actions. The analysis relies on the expected effect of the proximity of pylons on single-family house prices since it is based on the announcement period alone. To our knowledge, this is one of the first research papers to take advantage of such public decisions to investigate the impact on expectation. Second, the local application of the before/after investigation using a DiD hedonic pricing model allows us to adequately control for the omission of spatial variable bias, as well as controlling adequately for spatial autocorrelation within residuals. Finally, a global cost-benefit analysis to investigate how such public action can result in a net positive impact from an aggregated perspective reveals interesting features regarding the potential net impact of burying high voltage lines.

The analysis also has its limitations. First, it is based on expectations about the real-estate market, which could overestimate the final impact after a market adjustment process. It clearly shows that the presence of electricity transmission lines does entail a local cost that is borne by a few individuals. Second, the analysis relies on a relatively small number of transactions. This raises questions about the external validity of the estimation, as it is conducted in a specific region and in a limited number of municipalities. However, most analyses involving rural areas necessarily face such constraints since transactions there are sparse.

From a public policy perspective, while several authors (Mueller et al., 2019; Lienert et al., 2018) have shown that undergrounding is not a panacea for limiting protests by local

populations, or for modifying their perception of the risks, the results suggest that, in some situations, the undergrounding of transmission lines might prove a useful way to limit negative local perceptions. The results suggest that if the local population demands underground technology despite its cost, the global economic impact might be efficient from a social welfare standpoint, at least in some areas.

References

- Antanokis, J., Bendahan, S., Jacquart, P. & Lalive, R. (2010). On making causal claims: A review and recommendations, *The Leadership Quarterly*, 21: 1086-1120.
- Bateman, I. J., Day, B. H., Georgiou, S., & Lake, I. (2006). The aggregation of environmental benefit values: welfare measures, distance decay and total WTP. *Ecological Economics*, 60(2), 450-460.
- Battaglini, A., Komendantova, N., Brtnik, P. & Patt, A. (2012). Perception of barriers for expansion of electricity grids in the European Union, *Energy Policy*, 47: 254-259.
- Bishop, K.C., Kuminoff, N.V., Banzhaf, H.S., Boyle, K.J., von Gravenitz, K., Pope, J.C. & Timmins, C.D. (2020). Best practices for using hedonic property value models to measure willingness to pay for environmental quality. *Review of Environmental Economics and Policy*, 14(2).
- Boslett, A., Guilfoos, T. & Lang, C. (2016). Valuation of expectations: A hedonic study of shale gas development and New York's moratorium. *Journal of Environmental Economics and Management*, 77: 14-30.
- Cain, N.L. & Nelson, H.T. (2013). What drives opposition to high-voltage transmission lines? *Land Use Policy*, 33: 204-213.
- Ciupuliga, A.R. & Cuppen, E. (2013). The role of dialogue in fostering acceptance of transmission lines: The case of a France–Spain interconnection project. *Energy Policy*, 60: 224-233.
- Cohen, J., Moeltner, K., Reichl, J. & Schmidthaler, M. (2016). An empirical analysis of local opposition to new transmission lines across the EU-27, *Energy Journal*, 37(3):59-82.
- Cohen, J.J., Reichl, J. & Schmidthaler, M. (2014). Re-focussing research efforts on the public acceptance of energy infrastructure: A critical review. *Energy*, 76: 4-9.
- Colwell, P. (1990). Power Lines and Land Value. *Journal of Real Estate Research*, 5: 117-127.
- Des Rosiers, F. (2002). Power lines, visual encumbrance and house values: A microspatial approach to impact measurement, *Journal of Real Estate Research*, 23: 275-302.

585 Devaux, N., Dubé, J. & Apparicio, P. (2017). Anticipation and post-construction impact of a
586 metro extension on residential values: The case of Laval (Canada), 1995–2013. *Journal of*
587 *Transport Geography*, 62: 8-19.

588 ENTSO-e (2014). “Ten-year network development plan 2014”. Report: 493 pages.

589 Gibbons, S. (2015). Gone with the Wind: Valuing the visual impacts of wind turbines through
590 house prices. *Journal of Environmental Economics and Management*, 72: 177-196.

591 Greenstone, M. & Gayer, T. (2009). Quasi-experimental and experimental approaches to
592 environmental economics. *Journal of Environmental Economics and Management*, 57(1): 21-
593 44.

594 Hamilton, S.W. & Schwann, G.M. (1995). Do high voltage electric transmission lines affect
595 property value?, *Land Economics*, 71: 436-444.

596 Kuminoff, N.V., Parmeter, C.F. & Pope, J.C. (2010). Which hedonic models can we trust to
597 recover the marginal willingness to pay for environmental amenities? *Journal of*
598 *Environmental Economics and Management*, 60(3): 145-160.

599 Lienert, P., Sutterlin, B. & Siegrist, M. (2018). Public acceptance of high-voltage power lines:
600 The influence of information provision on undergrounding. *Energy Policy*, 112: 305-315.

601 Maddison, D., Ogier, R., & Beltrán, A. (2022). The Disamenity Impact of Solar Farms: A
602 Hedonic Analysis. *Land Economics*, 071220-0105R.

603 Menges, R., & Beyer, G. (2014). Underground cables versus overhead lines: Do cables
604 increase social acceptance of grid development? Results of a Contingent Valuation survey in
605 Germany. *International Journal of Sustainable Energy Planning and Management*, 3, 33-48.

606 Mueller, C.E., Keil, S.I. & Bauer, C. (2019). Underground cables vs. overhead lines: Quasi-
607 experimental evidence for the effects on public risk expectations, attitudes, and protest
608 behavior. *Energy Policy*, 125: 456-466.

609 Murray, J. & Bardaka, E. (2021). Evaluating the spatial and temporal distribution of beltway
610 effects on housing prices using difference-in-differences methods. *Transportation*, 1-36.

611 Navrud, S., Ready, E.C., Magnussen K. & Bergland, K, O. (2008). Valuing the social benefits
612 of avoiding landscape degradation from overhead power transmission lines: Do underground
613 cables pass the benefit–cost test?, *Landscape Research*, 33(3): 281-296.

614 Parmeter, C.F. & Pope, J.F. (2013). “Quasi-experiments and hedonic property value
615 Methods”, in J.A. List & M.K. Price (Eds), *Handbook on Experimental Economics and the*
616 *Environment*, Northampton, Ma: Edward Elgar Publishing.

617 Rious, V., Perez, Y. & Glachant, J.-M. (2011). Power transmission network investment as an
618 anticipation problem, *Review of Network Economics*, 10(4).

619 Rosen, S. (1974). Hedonic prices and implicit prices: Product differentiation in pure
620 competition, *Journal of Political Economy*, 82 (1): 34-35.

- 621 Sims, S. & Dent, P. (2005). High-voltage overhead power lines and property values: A
622 residential study in the UK, *Urban Studies*, 42: 665-694.
- 623 Soini, K., Pouta, E., Salmiovirta, M., Uusitalo, M. & Kivinen, T. (2011). Local residents'
624 perceptions of energy landscape: The case of transmission lines, *Land Use Policy*, 28: 294-
625 305.
- 626 Sun, C., Meng, X., Ouyang, X., & Xu, M. (2023). Social cost of waste-to-energy (WTE)
627 incineration siting: From the perspective of risk perception. *Environmental Impact*
628 *Assessment Review*, 102, 107204.
- 629 Tang, C.T. & Gibbons, S. (2021). Power lines and negative externalities: Evidence from the
630 housing market, 37 pages.
- 631 Tempesta, T., Vecchiato, D., & Girardi, P. (2014). The landscape benefits of the burial of
632 high voltage power lines: A study in rural areas of Italy. *Landscape and Urban Planning*, 126,
633 53-64.
- 634 Tiebout, C.M. (1956). The urban economic base reconsidered, *Land Economics*, 32(1): 95-99.
- 635 Wadley, D.A., Han, J.H., & Elliott, P.G. (2019). Risk hidden in plain sight: Explaining
636 homeowner perceptions of electricity transmission infrastructure, *Energy Policy*, 132: 744-
637 753.